抑郁与代谢综合征症状网络分析:基于 CHARLS 与 ELSA 的横断面研究

结果报告(Results) · 2026-06-18

本研究采用高斯图模型(GGM)结合 LASSO 正则化与 EBIC,分别构建中国健康与养老追踪调查(CHARLS)与英国老龄化纵向研究(ELSA)两个队列中抑郁症状(CESD)与代谢综合征(MetS)组分的症状网络,以 Fruchterman-Reingold 算法可视化,并计算节点强度(Strength)、中介(Betweenness)、接近(Closeness)中心性、桥梁中心性(Bridge EI / Bridge Strength)等指标;采用非参数 Bootstrap 评估网络稳定性与准确性,并以网络比较检验(NCT)比较两性的网络差异。两队列结果分别报告如下。

Table 1. Baseline characteristics of the study populations in CHARLS and ELSA.

变量	CHARLS (n = 8,412)	ELSA (n = 3,166)
年龄(岁),中位数 (IQR)	59.00 (53.00, 66.00)	65.00 (60.00, 71.00)
女性,n (%)	4,506 (53.57%)	1,808 (57.11%)
男性,n (%)	3,906 (46.43%)	1,358 (42.89%)
CESD 总分,均值 ± SD	8.50 ± 6.39	1.18 ± 1.81
抑郁患病率,n (%)	3,203 (38.08%)	541 (17.09%)
代谢综合征患病率,n (%)	3,404 (40.47%)	742 (23.44%)

如表 1 所示,CHARLS 队列年龄中位数为 59.00 岁、女性占 53.57%,CESD 总分 8.50 ± 6.39,抑郁与代谢综合征患病率分别为 38.08% 与 40.47%;ELSA 队列年龄中位数为 65.00 岁、女性占 57.11%,CESD 总分 1.18 ± 1.81,抑郁与代谢综合征患病率分别为 17.09% 与 23.44%。两队列均以女性略多,CHARLS 队列的抑郁与代谢综合征负担均明显高于 ELSA。

第一部分 · CHARLS 队列结果

Table 2. CHARLS:Node-level descriptive statistics of the depression-metabolic syndrome network (CESD-10 + 5 MetS components).

Label	Items	Mean	SD	Predictability (R^2)
Depression 1	Bothered by Things	1.035	1.122	0.382
Depression 2	Trouble Keeping Mind on Things	0.925	1.093	0.302
Depression 3	Felt Depressed	0.977	1.076	0.467
Depression 4	Everything Was an Effort	1.033	1.154	0.349
Depression 5	Hopeful About Future [POS]	1.158	1.211	0.179
Depression 6	Felt Fearful	0.352	0.785	0.207
Depression 7	Restless Sleep	1.051	1.193	0.159
Depression 8	Happy [POS]	1.058	1.124	0.288
Depression 9	Felt Lonely	0.547	0.961	0.278
Depression 10	Could Not Get Going	0.365	0.800	0.275
MetS 1	Waist Circumference (cm)	85.263	10.140	0.166
MetS 2	HDL-Cholesterol (mg/dL)	51.692	15.131	0.262
MetS 3	Triglycerides (mg/dL)	123.829	73.412	0.267
MetS 4	Systolic Blood Pressure (mmHg)	130.656	21.458	0.056
MetS 5	Fasting Glucose (mg/dL)	108.749	32.328	0.064

表 2 列出 CHARLS 网络 15 个节点(10 个 CESD-10 抑郁条目 + 5 个代谢综合征组分)的均值、标准差与可预测性。抑郁条目可预测性以 Depression 3(Felt Depressed, R^2=0.467)最高;代谢组分中收缩压(MetS 4, R^2=0.056)与空腹血糖(MetS 5, R^2=0.064)可预测性最低,提示其难以被网络中其余节点预测。甘油三酯(MetS 3)与空腹血糖的离散度较大(SD 分别为 73.41、32.33),反映其分布的明显右偏。

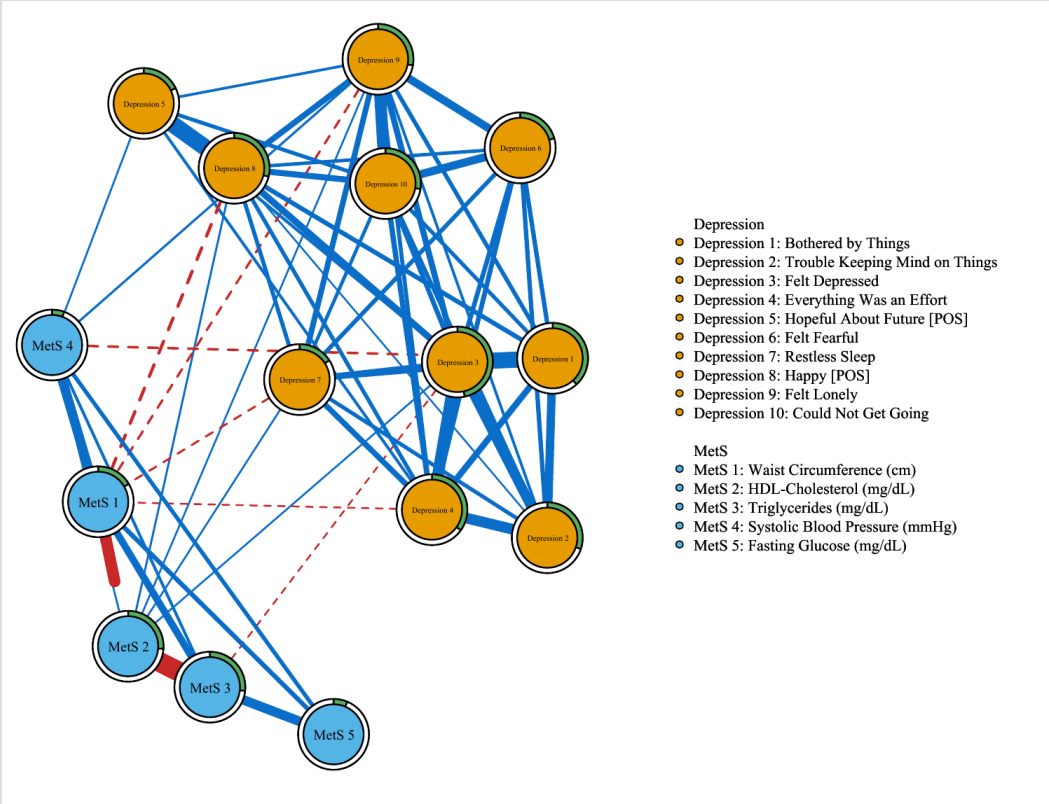


Figure 1. CHARLS 抑郁-代谢综合征症状网络。节点代表各症状/组分,边表示偏相关;蓝色实线为正相关,红色虚线为负相关,边宽对应相关强度。网络呈双簇结构:10 个抑郁条目形成致密的内部连接簇,5 个代谢组分形成较稀疏的簇,两簇间以较细的跨簇边相连。抑郁簇内最强边为 Depression 3(Felt Depressed)↔ Depression 1(Bothered by Things),其次为 Depression 3 ↔ Depression 4(Everything Was an Effort),以及两个正性条目 Depression 8(Happy)↔ Depression 5(Hopeful)。代谢簇内 MetS 2(HDL)与 MetS 1(腰围)、MetS 3(甘油三酯)之间为显著的红色粗线(强负相关),符合 HDL 的保护性方向。

中心性分析

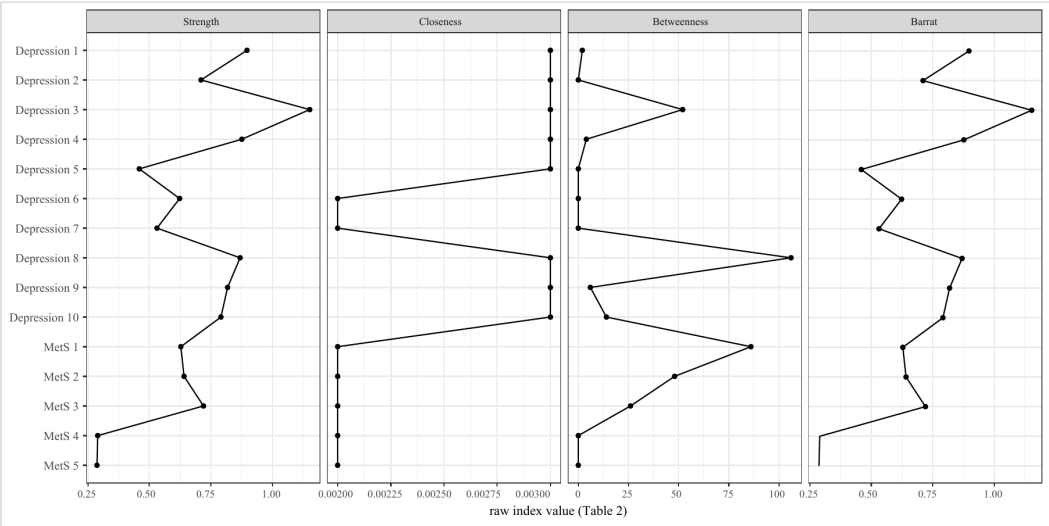


Figure 2. CHARLS 各节点的中心性指标(强度 Strength、接近 Closeness、中介 Betweenness、Barrat)。强度最高的节点为 Depression 3(Felt Depressed, 1.152),其后依次为 Depression 1(0.897)、Depression 4(0.876)、Depression 8(Happy, 0.869)与 Depression 9(Felt Lonely, 0.818);代谢组分中收缩压(MetS 4, 0.290)与空腹血糖(MetS 5, 0.287)强度最低。中介中心性以 Depression 8(Happy, 106)、MetS 1(腰围, 86)与 Depression 3(52)为高。

Table 3. CHARLS:Descriptive statistics of centrality indices (standardized z-values).

Label	Items	Strength	Closeness	Betweenness	Barrat	Onnela
Depression 1	Bothered by Things	0.897	0.003	2	0.897	0.174
Depression 2	Trouble Keeping Mind on Things	0.710	0.003	0	0.710	0.178
Depression 3	Felt Depressed	1.152	0.003	52	1.152	0.143
Depression 4	Everything Was an Effort	0.876	0.003	4	0.876	0.152
Depression 5	Hopeful About Future [POS]	0.459	0.003	0	0.459	0.112
Depression 6	Felt Fearful	0.623	0.002	0	0.623	0.190
Depression 7	Restless Sleep	0.531	0.002	0	0.531	0.115
Depression 8	Happy [POS]	0.869	0.003	106	0.869	0.103
Depression 9	Felt Lonely	0.818	0.003	6	0.818	0.102
Depression 10	Could Not Get Going	0.791	0.003	14	0.791	0.175
MetS 1	Waist Circumference (cm)	0.628	0.002	86	0.628	0.069
MetS 2	HDL-Cholesterol (mg/dL)	0.641	0.002	48	0.641	0.057
MetS 3	Triglycerides (mg/dL)	0.720	0.002	26	0.720	0.129
MetS 4	Systolic Blood Pressure (mmHg)	0.290	0.002	0	0.290	0.046
MetS 5	Fasting Glucose (mg/dL)	0.287	0.002	0	0.287	0.210

表 3 给出 CHARLS 各节点的标准化中心性指标。强度中心性显示 Depression 3(Felt Depressed)为网络核心节点,抑郁条目整体占据中心位置,而代谢组分(尤其收缩压、空腹血糖)处于边缘。接近中心性在所有节点间差异很小(0.002–0.003),全局连接较为均匀。Onnela 加权指标中 Depression 6(Felt Fearful, 0.190)与 MetS 5(空腹血糖, 0.210)相对较高,提示其在局部子结构中具一定重要性。

桥梁症状分析

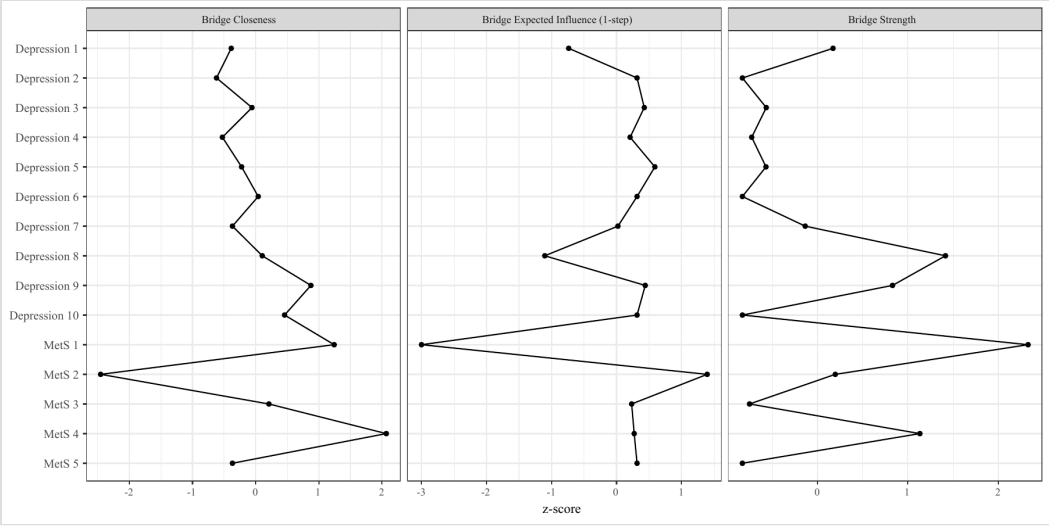


Figure 3. CHARLS 桥梁症状分析(Bridge Closeness、Bridge Expected Influence、Bridge Strength)。Bridge Expected Influence 最高者为 MetS 2(HDL, z≈1.4),抑郁条目中以 Depression 5(Hopeful, z≈0.6)较突出;Bridge Strength 最高者为 MetS 1(腰围, z≈2.4,全图最高),其次为 Depression 8(Happy, z≈1.4)与 MetS 4(收缩压, z≈1.1)。结果显示连接抑郁-代谢两簇的核心桥梁以代谢组分(腰围、HDL)为主。

性别差异

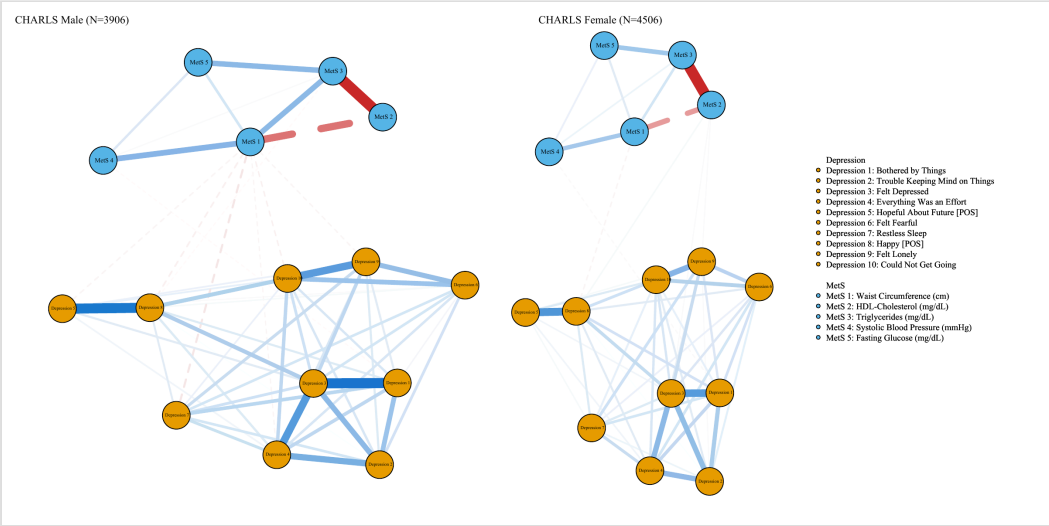


Figure 4. CHARLS 按性别分层的抑郁-代谢综合征网络(左:男性 n=3,906;右:女性 n=4,506)。Depression 3(Felt Depressed)在两性中均为强度最高的核心节点。NCT 检验显示两性网络**结构**存在显著差异(M=0.1054, P=0.002),但**全局强度**无显著差异(M=0.0498, P=0.732;男 4.95 vs 女 5.00);逐边检验有 7 条边、3 项中心性存在显著性别差异。

Table 4. CHARLS:Standardized centrality indices stratified by gender (closeness omitted as near-constant).

Label	Items	Male Strength	Male Between.	Female Strength	Female Between.
Depression 1	Bothered by Things	0.790	16	0.915	0
Depression 2	Trouble Keeping Mind on Things	0.698	0	0.687	0
Depression 3	Felt Depressed	1.144	26	1.130	38
Depression 4	Everything Was an Effort	0.852	2	0.886	8
Depression 5	Hopeful About Future [POS]	0.459	0	0.452	0
Depression 6	Felt Fearful	0.547	0	0.625	0
Depression 7	Restless Sleep	0.527	70	0.499	0
Depression 8	Happy [POS]	0.807	26	0.868	106
Depression 9	Felt Lonely	0.814	16	0.756	8
Depression 10	Could Not Get Going	0.801	4	0.771	0
MetS 1	Waist Circumference (cm)	0.725	86	0.511	86
MetS 2	HDL-Cholesterol (mg/dL)	0.566	0	0.677	48
MetS 3	Triglycerides (mg/dL)	0.668	26	0.721	26
MetS 4	Systolic Blood Pressure (mmHg)	0.226	0	0.232	0
MetS 5	Fasting Glucose (mg/dL)	0.269	0	0.261	0

表 4 显示,Depression 3(Felt Depressed)在男(1.144)女(1.130)两性中均为强度最高的核心节点。男性网络中 Depression 7(Restless Sleep)的中介中心性(70)较高,女性网络中 Depression 8(Happy)的中介中心性(106)最高。代谢组分在两性中的强度总体相近。

网络稳定性与准确性

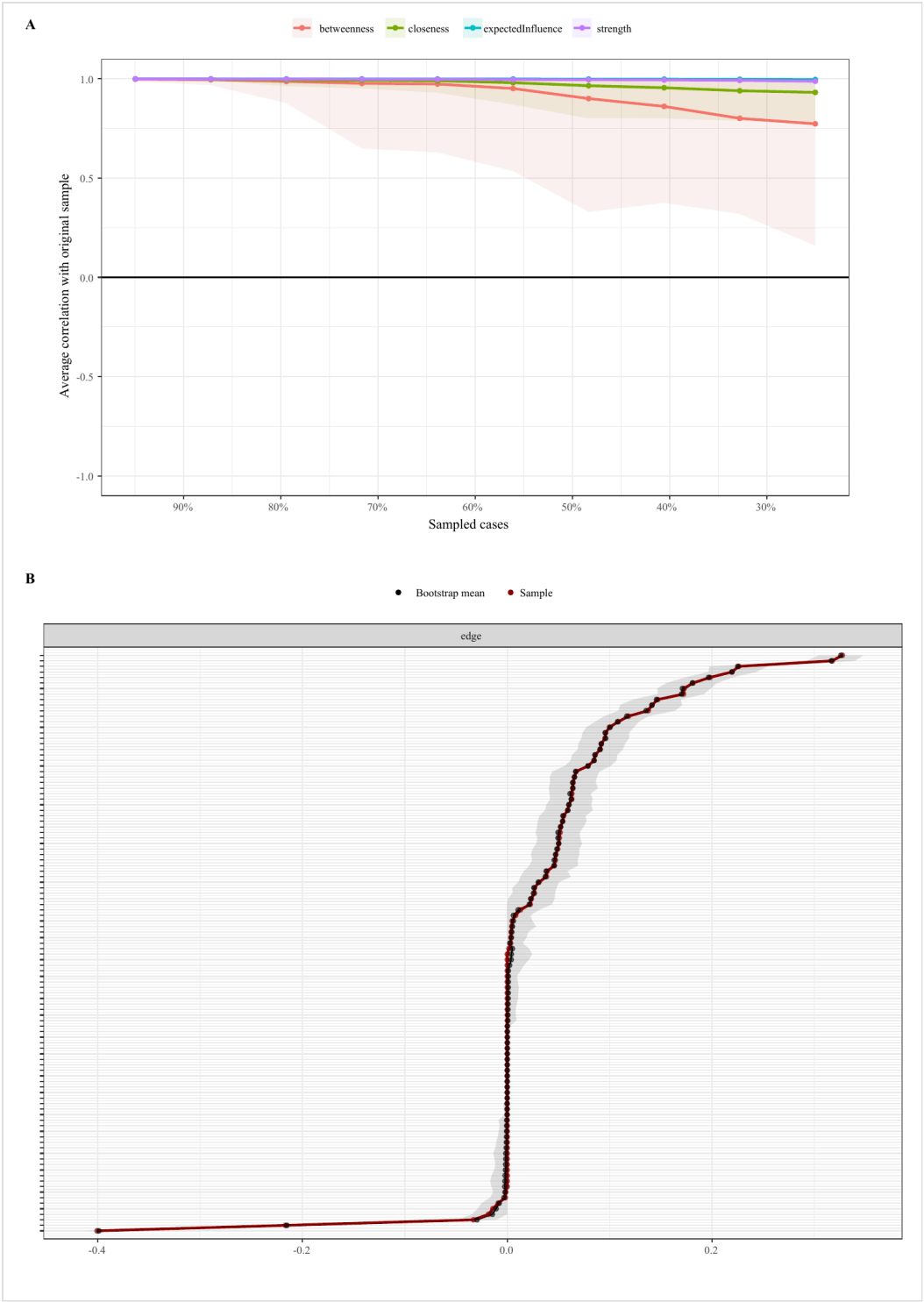


Figure 5. CHARLS 网络稳定性与准确性分析。A:中心性稳定性,相关稳定性系数(CS-coefficient, cor=0.7)分别为 Strength=0.75、Closeness=0.75、Expected Influence=0.75、Betweenness=0.361,四项均高于 0.25 阈值,提示各中心性指标稳定可信。B:边权重的 Bootstrap 准确性,95% 置信区间较窄,边权重估计准确。

第二部分 · ELSA 队列结果

Table 5. ELSA:Node-level descriptive statistics of the depression-metabolic syndrome network (CESD-8 + 5 MetS components).

Label	Items	Mean	SD	Predictability (R^2)
Depression 1	Felt Depressed	0.110	0.313	0.441
Depression 2	Everything Was an Effort	0.148	0.356	0.417
Depression 3	Restless Sleep	0.338	0.473	0.152
Depression 4	Happy [POS]	0.096	0.294	0.445
Depression 5	Felt Lonely	0.097	0.296	0.268
Depression 6	Enjoyed Life [POS]	0.081	0.273	0.458
Depression 7	Felt Sad	0.166	0.372	0.352
Depression 8	Could Not Get Going	0.146	0.353	0.382
MetS 1	Waist Circumference (cm)	94.786	13.471	0.267
MetS 2	HDL-Cholesterol (mg/dL)	65.512	18.387	0.331
MetS 3	Triglycerides (mg/dL)	120.932	65.130	0.244
MetS 4	Systolic Blood Pressure (mmHg)	131.875	17.067	0.038
MetS 5	Fasting Glucose (mg/dL)	97.080	17.164	0.055

表 5 列出 ELSA 网络 13 个节点(8 个 CESD-8 抑郁条目 + 5 个代谢组分)的描述统计。ELSA 采用 CESD-8,抑郁条目均值普遍较低(0.08–0.34);可预测性以 Depression 6(Enjoyed Life, R^2=0.458)最高,收缩压(MetS 4, R^2=0.038)与空腹血糖(MetS 5, R^2=0.055)最低,与 CHARLS 一致地提示血压、血糖难以被其余节点预测。

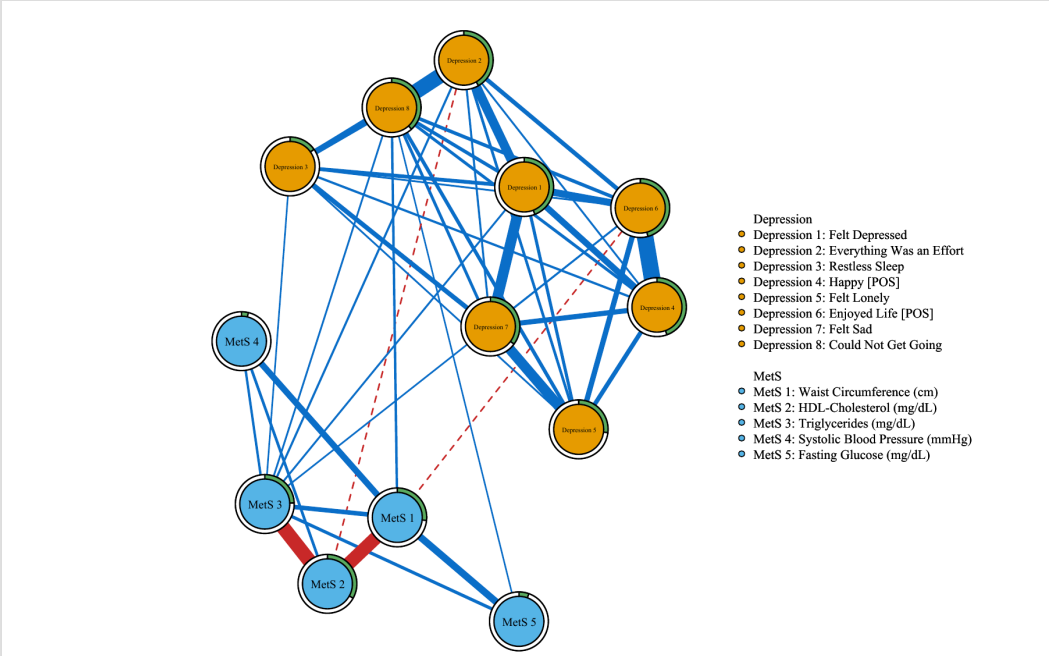


Figure 6. ELSA 抑郁-代谢综合征症状网络(说明同 Figure 1)。网络同样呈双簇结构。抑郁簇内最强边为 Depression 1(Felt Depressed)↔ Depression 7(Felt Sad),其次为两个正性条目 Depression 4(Happy)↔ Depression 6(Enjoyed Life),以及 Depression 2(Everything Was an Effort)↔ Depression 8(Could Not Going)。代谢簇内 MetS 2(HDL)与 MetS 1(腰围)、MetS 3(甘油三酯)之间同样为显著的红色粗线(强负相关)。

中心性分析

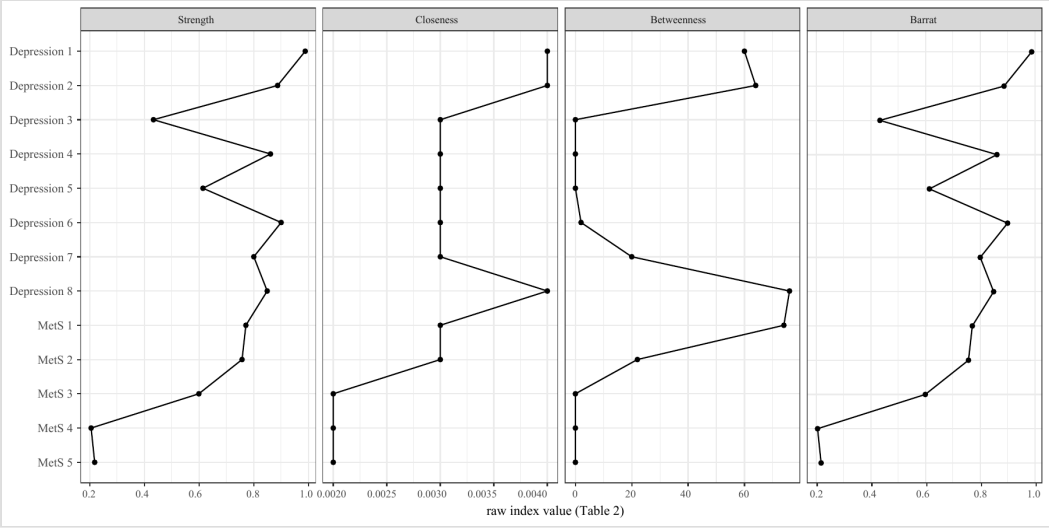


Figure 7. ELSA 各节点的中心性指标。强度最高者为 Depression 1(Felt Depressed, 0.988),其后为 Depression 6(Enjoyed Life, 0.900)、Depression 2(0.887)与 Depression 4(Happy, 0.861);代谢组分中收缩压(MetS 4, 0.205)与空腹血糖(MetS 5, 0.218)强度最低。中介中心性以 Depression 8(Could Not Get Going, 76)、MetS 1(腰围, 74)与 Depression 2(64)为高;**但 ELSA 中介中心性的稳定性不足(CS=0.05,见 Figure 10),该指标结果应谨慎解读,结论以 Strength 为主。**

Table 6. ELSA:Descriptive statistics of centrality indices (standardized z-values).

Label	Items	Strength	Closeness	Betweenness	Barrat	Onnela
Depression 1	Felt Depressed	0.988	0.004	60	0.988	0.180
Depression 2	Everything Was an Effort	0.887	0.004	64	0.887	0.115
Depression 3	Restless Sleep	0.433	0.003	0	0.433	0.107
Depression 4	Happy [POS]	0.861	0.003	0	0.861	0.185
Depression 5	Felt Lonely	0.614	0.003	0	0.614	0.165
Depression 6	Enjoyed Life [POS]	0.900	0.003	2	0.900	0.114
Depression 7	Felt Sad	0.800	0.003	20	0.800	0.177
Depression 8	Could Not Get Going	0.849	0.004	76	0.849	0.097
MetS 1	Waist Circumference (cm)	0.771	0.003	74	0.771	0.091
MetS 2	HDL-Cholesterol (mg/dL)	0.757	0.003	22	0.757	0.172
MetS 3	Triglycerides (mg/dL)	0.599	0.002	0	0.599	0.051
MetS 4	Systolic Blood Pressure (mmHg)	0.205	0.002	0	0.205	0.201
MetS 5	Fasting Glucose (mg/dL)	0.218	0.002	0	0.218	0.100

表 6 给出 ELSA 各节点的标准化中心性指标。与 CHARLS 类似,Depression 1(Felt Depressed)为网络核心节点,代谢组分(尤其收缩压、空腹血糖)处于边缘;接近中心性在所有节点间差异较小(0.002–0.004)。

桥梁症状分析

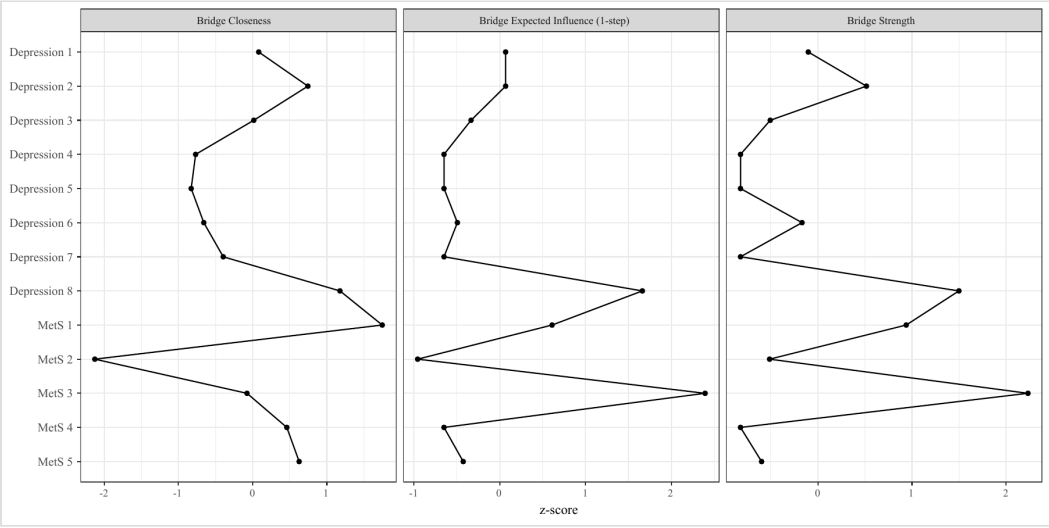


Figure 8. ELSA 桥梁症状分析。Bridge Expected Influence 最高者为 MetS 3(甘油三酯, z≈2.4),抑郁条目中以 Depression 8(Could Not Get Going, z≈1.7)最高;Bridge Strength 同样以 MetS 3(甘油三酯, z≈2.2)最高,其次为 Depression 8(z≈1.5)与 MetS 1(腰围, z≈1.0)。在 ELSA 中,甘油三酯是连接抑郁-代谢两簇的最主要桥梁组分。

性别差异

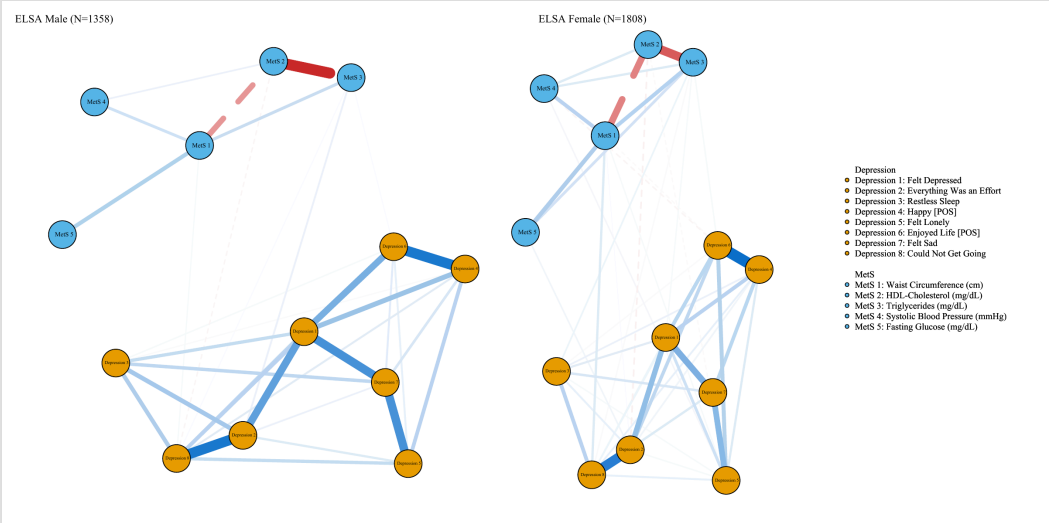


Figure 9. ELSA 按性别分层的抑郁-代谢综合征网络(左:男性 n=1,358;右:女性 n=1,808)。Depression 1(Felt Depressed)在男性中为强度最高的核心节点(1.075);女性中则以 Depression 6(Enjoyed Life, 0.984)强度最高,Depression 1(0.952)次之;代谢组分在女性中强度普遍更高。NCT 检验显示两性网络**结构**无显著差异(M=0.1357, P=0.156),但**全局强度**显著不同(M=0.8005, P=0.002),女性(4.71)高于男性(3.91);逐边检验有 15 条边、7 项中心性存在显著性别差异。

Table 7. ELSA:Standardized centrality indices stratified by gender (closeness omitted as near-constant).

Label	Items	Male Strength	Male Between.	Female Strength	Female Between.
Depression 1	Felt Depressed	1.075	60	0.952	28
Depression 2	Everything Was an Effort	0.839	74	0.921	66
Depression 3	Restless Sleep	0.462	0	0.411	0
Depression 4	Happy [POS]	0.720	0	0.930	2
Depression 5	Felt Lonely	0.566	0	0.662	0
Depression 6	Enjoyed Life [POS]	0.758	2	0.984	18
Depression 7	Felt Sad	0.790	18	0.776	18
Depression 8	Could Not Get Going	0.836	0	0.864	78
MetS 1	Waist Circumference (cm)	0.458	42	0.812	72
MetS 2	HDL-Cholesterol (mg/dL)	0.595	54	0.768	20
MetS 3	Triglycerides (mg/dL)	0.492	64	0.748	0
MetS 4	Systolic Blood Pressure (mmHg)	0.103	0	0.311	0
MetS 5	Fasting Glucose (mg/dL)	0.117	0	0.273	0

表 7 显示,Depression 1(Felt Depressed)在男性中为强度最高的核心节点(1.075);而在女性网络中,Depression 6(Enjoyed Life, 0.984)强度最高,Depression 1(0.952)紧随其后。值得注意的是,代谢组分在女性中强度普遍更高(如腰围:男 0.458 vs 女 0.812;HDL:男 0.595 vs 女 0.768;甘油三酯:男 0.492 vs 女 0.748),提示代谢簇在女性网络中连接更紧密,与 NCT 显示的女性全局强度更高相吻合。

网络稳定性与准确性

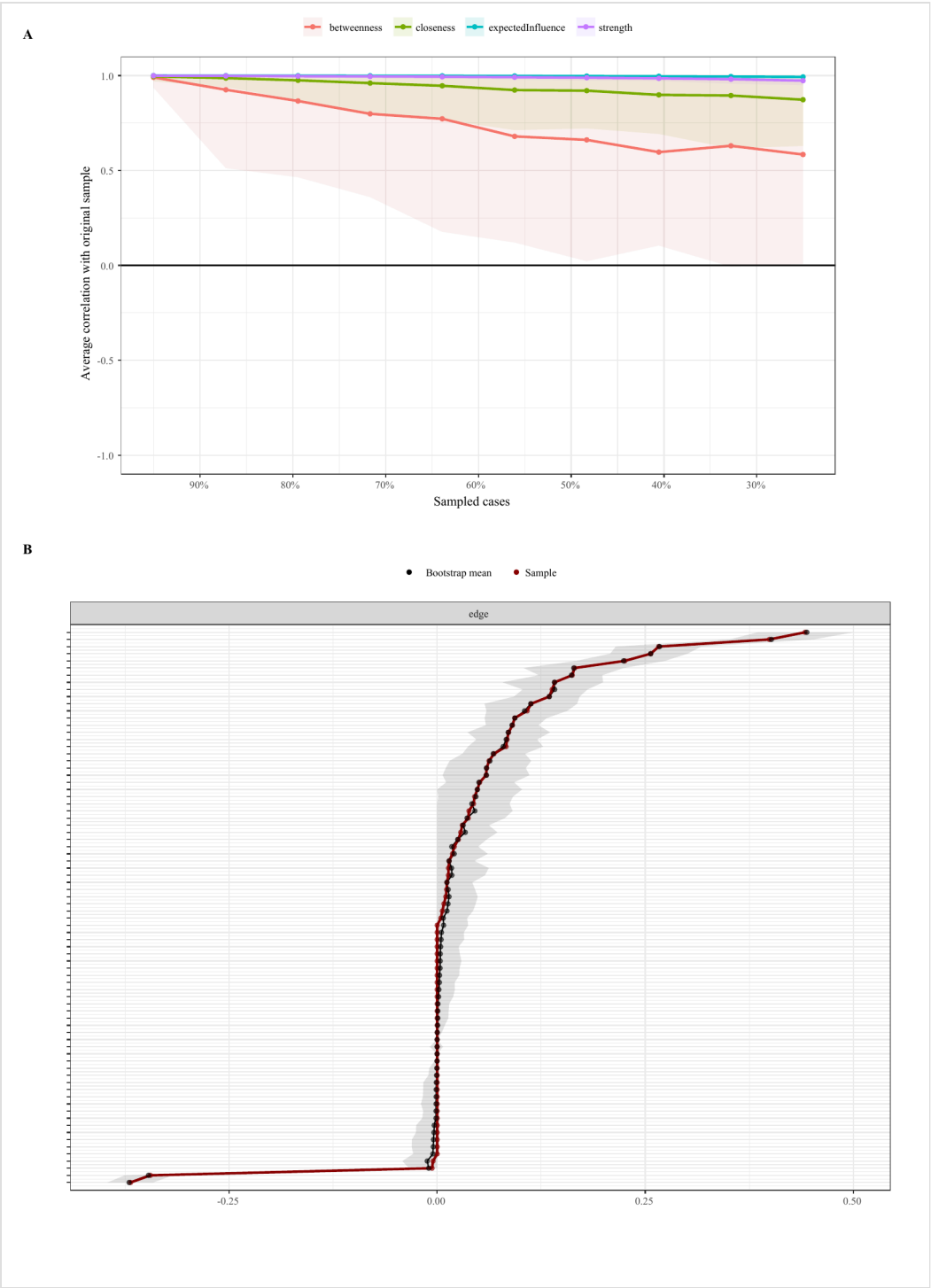


Figure 10. ELSA 网络稳定性与准确性分析。A:中心性稳定性,相关稳定性系数(CS-coefficient, cor=0.7)分别为 Strength=0.75、Closeness=0.594、Expected Influence=0.75,均稳定;但**Betweenness=0.05(<0.25 阈值,不稳定)**,故 ELSA 的中介中心性结果应谨慎解读,网络解读以 Strength 为主。B:边权重 Bootstrap 准确性,95% 置信区间较窄,边权重估计准确。